LESSON 10.4

Math is Ingenuity

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use ratios and models to explain how the Penny Farthing and gear-driven bicycles were ingenious solutions to a real problem.

WHY THIS MATTERS The invention of the bicycle in the late 1800s offered people a way to move about easily. Among the first bicycles created was the Penny Farthing, with a large front wheel and a small rear wheel. Ingenuity in mathematics helps us come up with different, and sometimes unique, solutions to problems.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
ingenuity *Spanish: ingenio*	Coming up with different, and sometimes unique, solutions to problems.	The Penny Farthing's huge wheel, then chains and gears — two clever fixes.
rotation *Spanish: rotación*	One full turn of a wheel, gear, or pedal all the way around.	One full turn of the pedal — 360° all the way around.
gear *Spanish: engranaje*	A toothed wheel that passes motion to another toothed wheel through a chain.	A 48-tooth pedal gear pulling a 12-tooth wheel gear through the chain.
ratio *Spanish: razón*	A comparison of two quantities, like pedal rotations to wheel rotations.	48 pedal teeth to 12 wheel teeth — 4 wheel turns for every pedal turn.
equivalent ratios *Spanish: razones equivalentes*	Ratios that make the same comparison using different numbers.	1 : 4 and 3 : 12 describe the same gear, at different sizes.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A later bicycle design uses a chain: the pedal gear has 48 teeth, and the largest rear gear has 32 teeth.

1. For each rotation of the 48-tooth pedal gear, the wheel gear also turns 48 teeth.
2. The 32-tooth gear needs to turn 32 teeth to make one full rotation, so I find the equivalent ratio: 48 ÷ 32 = 1.5.
3. The rear wheel makes 1.5 rotations for each pedal rotation — this gear makes it easier to pedal at slower speeds or go uphill.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

The smallest rear gear has 12 teeth, and the pedal gear still turns 48 teeth per rotation.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- With the _____-tooth gear the ratio is _____, so the wheel _____.
- A bigger wheel travels _____ with each pedal rotation because _____ .
- A smaller wheel would travel _____ .
- One ingenious solution is _____ .

Word bank ingenuity | rotation | gear | ratio | equivalent ratios

WATCH OUT!

A common mistake

Thinking a rear gear with MORE teeth makes the wheel spin more

LESSON 10.4

Math is Ingenuity

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

On the Penny Farthing, what is the ratio of pedal rotations to front-wheel rotations?

(A) 1 to 1
(B) 1 to 2
(C) 1 to 4
(D) 2 to 1

HINT The pedals are attached directly to the front wheel.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

How far does the Penny Farthing travel with each pedal rotation?

(A) The height of the rider
(B) The length of the perimeter of the front wheel
(C) The length of the pedal
(D) It does not move a fixed distance

HINT Picture the wheel rolling one full turn on the ground.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Why was the front wheel of the Penny Farthing so large?

(A) So the bicycle is easier to stop

(B) So the rider sits closer to the ground

(C) So the bicycle travels a greater distance with one pedal rotation

(D) So the rear wheel spins faster

HINT Each pedal rotation moves the bike one perimeter of the wheel.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

For each rotation of the 48-tooth pedal gear, how many teeth does the wheel gear turn?

(A) 1 tooth

(B) 12 teeth

(C) 32 teeth

(D) 48 teeth

HINT The chain moves one link for each tooth on the pedal gear.

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

How does the size of the front wheel affect the distance the bicycle travels?

SENTENCE STARTER A bigger wheel travels ___ with each pedal rotation because ___ .
A smaller wheel would travel ___ .

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

Name one ingenious solution — from this lesson or from your own life — where math helped solve an everyday problem.

SENTENCE STARTER One ingenious solution is ___ .
The math behind it is ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Your partner says the bigger rear gear makes the wheel turn more. Is that right, and what stays equivalent in the ratio of teeth when you switch gears?

SENTENCE STARTER With the ___-tooth gear the ratio is ___, so the wheel ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 10.4

Math is Ingenuity

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

The pedal gear turns 48 teeth per rotation. The largest rear gear needs 32 teeth for one full rotation. How many rotations does the rear wheel make for each pedal rotation?

(A) 0.5 rotations
(B) 1.5 rotations
(C) 2 rotations
(D) 16 rotations

PER 1

2 **Circle the letter of the best answer. Show how you know.**

The smallest rear gear has 12 teeth. How many rotations will the rear wheel make for each rotation of the 48-tooth pedal gear?

(A) 0.25 rotations
(B) 3 rotations
(C) 4 rotations
(D) 36 rotations

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

You are about to ride up a steep hill. Which rear gear should the chain be on, and why?

- (A) The 12-tooth gear, because 4 rotations per pedal is always better
- (B) The 32-tooth gear, because it makes the bike go faster
- (C) It does not matter which gear you use on a hill
- (D) The 32-tooth gear, because 1.5 rotations per pedal makes pedaling easier at slow speeds

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

With the chain on the 12-tooth gear, the rider makes 10 pedal rotations. How many rotations does the rear wheel make?

- (A) 14 rotations
- (B) 22 rotations
- (C) 40 rotations
- (D) 120 rotations

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

Why do you think the larger gear makes it easier to pedal at slower speeds?

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

When the chain is on the largest gear the rear wheel makes 1.5 rotations per pedal rotation; on the smallest gear it makes 4. What do you notice about the size of the wheel gears and the distance the bike travels?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Your partner says the bigger rear gear makes the wheel turn more. Is that right, and what stays equivalent in the ratio of teeth when you switch gears?

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Math is Ingenuity

Challenge · Extension

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Distance = rotations × circumference. Match each ride to its distance.

_____ 4 wheel rotations, 7 ft circumference	(A) 28 feet
_____ 3 wheel rotations, 6 ft circumference	(B) 18 feet
_____ 2 wheel rotations, 7 ft circumference	(C) 14 feet
_____ 48-tooth pedal gear with a 12-tooth rear gear	(D) 4 rotations per pedal turn

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

The rider makes 3 full pedal rotations with the chain on the 32-tooth gear. How many rotations does the rear wheel make?

(A) 3 rotations
(B) 4.5 rotations
(C) 6 rotations
(D) 96 rotations

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A designer adds a middle gear so the rear wheel makes exactly 2 rotations per rotation of the 48-tooth pedal gear. How many teeth must the new gear have?

(A) 16 teeth

(B) 22 teeth

(C) 24 teeth

(D) 96 teeth

PER 1

PART 3 Explain and correct

reasoning

4 **Answer in complete sentences.**

How would the distance traveled differ with a smaller front wheel on the Penny Farthing? Use the 1 to 1 pedal ratio in your answer.

WRITE YOUR ANSWER

5 **Answer in complete sentences.**

How did math help the bicycle's designers come up with ingenious solutions — first with the Penny Farthing, then with chains and gears?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Your partner says the bigger rear gear makes the wheel turn more. Is that right, and what stays equivalent in the ratio of teeth when you switch gears?

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